

An alternative view of unimolecular reaction theory

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A primitive form of unimolecular reaction theory is constructed, based on the master equation. Application of the theory depends upon the way in which eigenvalues of large stochastic matrices behave as certain of the matrix elements are varied. Some of the required eigenvalue properties have been deduced by numerical experiment, and others are available from earlier work on diatomic dissociation reactions. Comparison of the results from this primitive approach with those of existing theories of unimolecular reactions suggests some reinterpretation of parameters used in those theories: most interestingly, the number of effective oscillators occurring in Kassel theory (or its Slater-theory equivalent) appears to be the ratio, at infinite pressure, of the total rate to the rate of the only reaction channel which survives at the low-pressure limit.

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On a construit une forme primitive de la théorie des réactions unimoléculaires qui est basée sur l'équation maîtresse. L'application de la théorie dépend de la façon de se comporter des valeurs de eigen de grandes matrices stochastiques lorsque l'on fait varier certains éléments de la matrice. Faisant appel à des expériences numériques, on a déduit quelques unes des propriétés des valeurs de eigen qui sont requises; d'autres sont disponibles à partir de travaux antérieurs sur des réactions de dissociation diatomique. Une comparaison des résultats obtenus grâce à cette approche primitive avec ceux prévus par les théories de réaction unimoléculaire existante suggère que l'on doit faire quelques réinterprétations des paramètres utilisés dans ces théories: la réinterprétation la plus intéressante a trait au nombre d'oscillateurs effectifs se produisant dans la théorie de Kassel (ou le nombre équivalent dans la théorie de Slater) qui semble être le rapport, à pression infinie, de la vitesse totale par rapport à la vitesse du seul chemin réactionnel qui survie à la limite de basse pression.

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Introduction

The formulation of the rate of a unimolecular reaction using the master equation is relatively commonplace (1-3). However, despite this general acceptance of its validity, the approach has yielded singularly little in the nature of a concrete predictive theory. The basic reason for this is that in the master-equation approach, there is one differential equation for every discrete state of the molecule, and since an interesting molecule from the point of view of unimolecular reaction theory may have many millions of energy levels, one is faced with a two-fold problem, that of finding solutions for enormously large sets of simultaneous differential equations, whose constant coefficients comprise an even larger number of unknown transition rates between the discrete states (4). The usual method of trying to overcome these difficulties is to partition the set of energy levels into blocks whose size is determined by the average amount of internal energy per collision transferred to the heat bath (2); this is really no more than simu-

lating the real process using a model consisting of (at most) a few hundred levels, which is perfectly acceptable provided (4) that the partitioning is not too restrictive. We then encounter the main obstruction to the formulation of a useful theory of unimolecular reactions, since these sets of simultaneous differential equations, although transformed into matrix notation (2, 4), have to be solved numerically. Unfortunately, there appear to be no useful theorems in existence connecting the eigenvalues with the elements of these transition matrices (5, 6), even for the simplest conceivable changes in the elements. Modern developments in linear algebra, stimulated by complementary computational activity, will clearly lead to some new theorems in this area (7, 8), but it has to be remembered here that in our case, we are interested in the behaviour of the eigenvalue of smallest numerical magnitude, which is the rate constant, rather than in the dominant eigenvalue which is generally more amenable to study.

The aim of this investigation, therefore, was to

set out the simplest and most primitive master-equation formulation which exhibits unimolecular reaction characteristics. The behaviour of the eigenvalues was then examined numerically over a wide range of conditions using matrices of the order of about 30; a few calculations were carried out on much larger matrices, of the order of 300, to confirm that some of the properties of interest were not artefacts of the relatively small size of the transition-rate matrix. It is then conjectured that any 'theorems' so deduced will be valid for the almost infinitely large matrices required for a proper description of the unimolecular reaction problem. The supposition that progress can be made in this way is not entirely without foundation. Earlier work on diatomic dissociation reveals two things: first, that the eigenvalue representing the rate of the reaction is very insensitive to the transition probabilities assumed between the internal states of the molecule, even for systems consisting of quite moderately few energy levels (4, 9); and second, that once one considers the molecule to consist of more than 20 levels or so, the rigid one-to-one dependence (9-11) of certain eigenvalues (relating to the internal relaxation) on specific elements of the matrix disappears, and we enter a regime where network or connectivity effects seem to dominate, with the eigenvalues determined co-operatively by contributions from most of the elements of the matrix (9, 12, 13). Clearly, even in the simplest cases, any connection between eigenvalues and the elements in the matrix is extremely complicated (11) and equally clearly, one can hope to elucidate some of the rules without detailed knowledge of individual matrix elements, or transition probabilities.

Examination of Four Primitive Models

We will consider the simplest possible case, that of a molecule A, infinitely diluted in a heat bath of molecules M, decomposing to give products. The energy levels ϵ_i of the molecule are assumed to be slightly convergent as the energy increases, and the degeneracy g_i of each level is assumed, for the time being, to be unity. We have three choices for the possible description of the reaction process itself. (a) To assume a cut-off at some level which is assumed to be an absorbing barrier (1); this is unnecessarily restrictive, and precludes any discussion of the behaviour of states above the critical level, or of phenomena

like chemical activation or the decomposition of ions. (b) Following Widom (14) we can restrict ourselves to an isomerisation and include all the internal states of the product molecule as well: this gives, in principle, a totally complete and exact description of the process. However, most unimolecular reactions are studied experimentally far away from equilibrium, *e.g.* methyl isocyanide, cyclopropane, and obviously we can (for now) dispense with the luxury of including all the product states, with a consequent reduction in both the size and complexity of the problem; when we need to include any reference to the product molecules, we can simply treat them as structureless entities. (c) We can treat the reaction processes as a series of predissociations from certain energetic states of the molecule: if these predissociating levels are embedded in the total manifold of energy levels, then it is clear that the description can encompass, if necessary, such concepts as chemical activation or decomposition to more than one set of products.

It becomes clear that the present treatment has many similarities to those examined by Valence and Schlag (15) and by Tardy and Rabinovitch (16) a decade ago, but whereas they were forced to circumvent the exact solution of the master-equation problem because of computational limitations, we will go directly for the exact solution. The strategy is that we may be able to discover the behaviour of interest for unimolecular reactions using the exact solutions for relatively small matrices and then make a 'mental extrapolation' to those larger matrices of real interest. It should also be noted that there are strong similarities to some aspects of the papers of Buff and Wilson (17) and of Gilbert and Ross (18).

Figure 1a depicts the simplest model of this kind which exhibits unimolecular reaction characteristics. In the actual numerical experiments, a series of $N = 30$ internal energy levels was used, spread over an energy range of 100 kcal/mol: they were slightly convergent, with a quadratic dependence on the index i which would cause the levels to converge at 1000 kcal/mol; in a sense then, the latter figure can be regarded as the heat of atomisation of the molecule, but because the temperature is low, we can truncate the series of levels at a much lower energy value than this. There is one predissociating state, with a decay rate constant k_r , arbitrarily assigned as 10^{13} s^{-1} : in this scheme, this is the 20th level, placed at

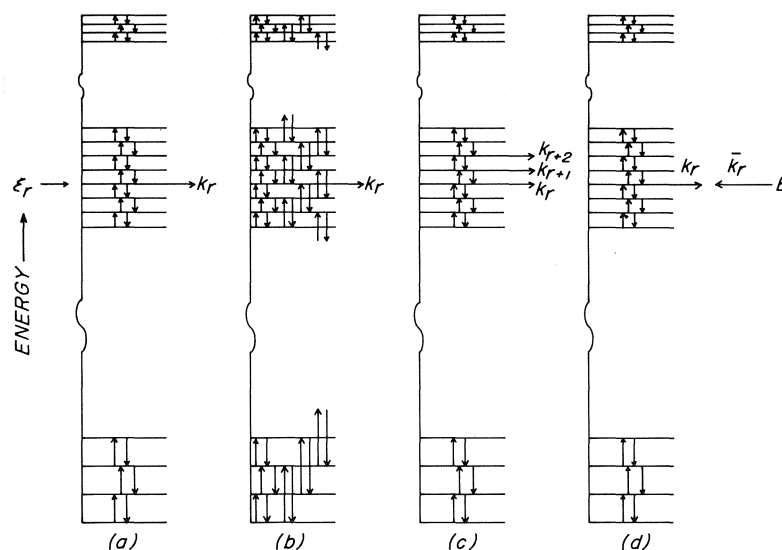


FIG. 1. Schematic representation of four unimolecular reaction models. All models have the same slightly convergent set of energy levels, and the same critical energy ϵ_r , but with the following variations: (a) nearest-neighbour-only transitions, dissipative, one predissociating level (model I); (b) one- and two-quantum transitions, dissipative, one predissociating level (model II); (c) nearest-neighbour-only transitions, dissipative, two or three predissociating levels (model III); (d) nearest-neighbour-only transitions, conservative, one predissociating level giving product molecule B; back-reaction rate constant $\bar{k}_r$ (model IV).

about $\epsilon = 70$ kcal/mol above the ground state; (note that the subscript r will be reserved for the index i when the state is a predissociating one). Each downward transition was given a transition probability which increased very weakly as the energy gap decreased, and all downward transitions had probabilities of the order of 10^{-3} per collision; upward transition probabilities were generated by detailed balancing from the equilibrium populations $\tilde{n}_i$ for the temperature in question. The temperature T was chosen in the range 500–1000 K to match the practical region in which a reaction with a critical energy of 70 kcal/mol might be studied. The actual numerical details of the model are immaterial, and are only presented for the purposes of clarifying the nature of the model, and as a vehicle for appreciating some of the later discussion; moreover, the philosophy of the experiment is that the model only serves to elucidate some numerical ‘theorems’, which can then be transferred to more realistic matrices of much larger size.

Most of the numerical experiments were performed upon what may be termed a dissipative model: in these calculations, the product molecules are unspecified and no reverse reaction

is included; such dissipative models are models I, II, and III, Figs. 1 *a*, *b*, and *c*. Representing the set of simultaneous differential equations in matrix form, the matrix elements for the dissipative cases are (13)

$$[1] \quad [A]_{ij} \equiv (1 - \delta_{ij})Q_{ji} - \delta_{ij} \left[k_i + \sum_{k=0}^{N-1} (1 - \delta_{ik})Q_{ki} \right]$$

where

$$Q_{ji} = [M]ZP_{ji} \\ \tilde{n}_i P_{ji} = \tilde{n}_j P_{ij}$$

and $[M]$ is the concentration of inert molecules, Z is the collision number between M and A , and P_{ji} is the probability per collision of a change in the internal state of A from $i \rightarrow j$. The matrix A_{ij} has N eigenvalues, none of which is zero because the system does not conserve particles: $(N - 1)$ of the eigenvalues represent the internal relaxation times ($\tau_j = -\lambda_j^{-1}$) and the eigenvalue of smallest numerical magnitude λ_N is (minus) the rate constant (k_{uni}) for the loss of particles from the system. Figure 2 shows the behaviour of these N eigenvalues, for fixed P_{ij} and fixed k_r , as a function of the inert gas concentration $[M]$: such

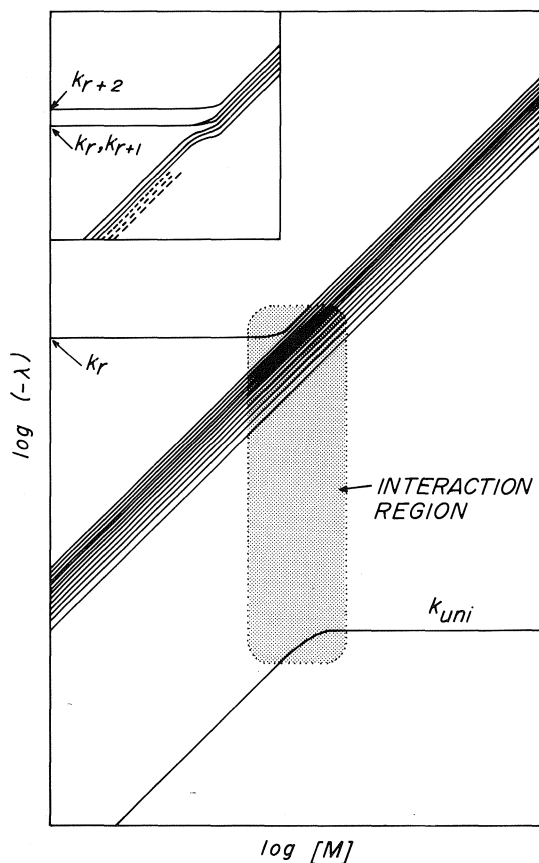


FIG. 2. Schematic representation of the behaviour of the eigenvalues of the reaction-rate matrices as a function of the concentration of M. The inset shows the behaviour of the six eigenvalues of largest numerical magnitude in the interaction region when there are three decay channels, two of them having the same decay constant *i.e.* $k_r = k_{r+1} \neq k_{r+2}$.

behaviour has been shown before for a two-state system (19), and this is the generalisation to an N -state system. The principal features are as follows: There is a bundle of $(N - 2)$ eigenvalues (very closely spaced because all the downward P_{ij} are very similar in numerical magnitude) which appear to be directly proportional to $[M]$; these are the rates of the internal relaxations of the molecule A in the presence of M. At very low $[M]$, there is an eigenvalue $\lambda_0 = 10^{13} \text{ s}^{-1}$, since this is much faster than any internal process under these conditions; however, as $[M]$ increases and the rates of the internal processes become comparable with 10^{13} s^{-1} , the largest eigenvalue is assimilated into the bundle. In the diagram, the bundle of internal eigenvalues is

drawn as though they were all straight lines, but in fact, in the interaction zone (depicted by the shaded area) each member of the bundle exhibits a small displacement such that in the limit of high pressure, λ_j is the continuation of the low-pressure λ_{j-1} (see the inset in Fig. 2 for more details, remembering that the inset actually refers to another calculation of type III); also, for each successive eigenvalue, the centre of the displacement moves slightly to the right as λ decreases in numerical magnitude. Finally, the eigenvalue of smallest numerical magnitude ($\lambda_N = -k_{\text{uni}}$) is parallel to the main bundle of internal-relaxation eigenvalues until the interaction region is reached, whence it turns over and becomes independent of $[M]$.

The properties of k_{uni} are as follows:

(a) *High pressure*

The infinite pressure rate constant is exactly

$$[2] \quad k_{\text{uni},\infty} = \bar{n}_r k_r; \quad \left(\text{normalisation } \sum_i n_i = 1 \right)$$

i.e. the product of the equilibrium population and the predissociation rate constant; this equation also remains true if the state $i = r$ is assigned a degeneracy g_r other than unity.

(b) *Low pressure*

The identification of the rate constant is not so decisive, and it appears that the low-pressure limiting form is

$$[3] \quad k_{\text{uni},0} \sim 0.26 \bar{n}_r |\lambda_{N-1}|$$

Obviously, this equation could be written in other suggestive forms, *e.g.* in terms of $\bar{n}_{r-1}$ which is the state supplying the predissociating one; other possible forms investigated involved the arithmetic or geometric means of the bundle of λ_j , but the relationships here were dependent on the number of levels considered, whereas in the form of [3], it was independent of both the number of levels and the index of the predissociating state, provided $r > \sim 5$, $N > 30$. Equally obviously, the rate of dissipation is simply $n_r k_r$, where n_r is the actual nonequilibrium population of the predissociating state, whence

$$[4] \quad \frac{n_r}{\bar{n}_r} \sim 0.26 \frac{|\lambda_{N-1}|}{k_r}$$

The coefficient 0.26 is clearly only appropriate for the particular set of energy levels and transition probabilities used in this calculation.

(c) Intermediate pressures

The most interesting property of k_{uni} is that it is exactly a Lindemann fall-off curve: close to a hundred values of k_{uni} as functions of $[M]$ have been obtained during the variety of numerical experiments carried out and in no case, given the high-pressure limit (eq. 2) and the numerical form of the low-pressure limit (eq. 3), did a value of k_{uni} for model I deviate by more than 1 part in 10^6 from Lindemann behaviour, *viz.*

$$[5] \quad k_{\text{uni}} = \frac{a[M]}{b + c[M]}$$

It seems safe to presume therefore that eq. 5 can be regarded as a valid rule for λ_N for large matrices of the form [1] with only one dissipative element k_r , given that r is embedded well within the body of the values of the energy-ordering index i ; eq. 2 identifies the ratio a/c unambiguously in terms of the properties of the matrix, and eq. 3 gives an approximate form for the ratio $a[M]/b$.

The next interesting question is whether this Lindemann behaviour is specific only to this primitive formulation, or whether it is more general: model I contains two exceedingly restrictive assumptions, that of nearest-neighbour-only transitions, and that of only one allowed predissociating state, and the Lindemann form for λ_N could be confined only to this special case.

The step-ladder or nearest-neighbour approximation (*viz.* $P_{ij} = 0, |i - j| \neq 1$) is a common one in theoretical chemical kinetics, and it is now well established (4, 10) that in diatomic dissociation, the activation flux is essentially step-ladder in form and that the principal result of making the step-ladder approximation is to effect a relatively small decrease in the overall reaction rate. Consequently we will examine this restriction first. This is done using model II, Fig. 1b, which includes both one- and two-quantum level transitions: the nearest-neighbour transition probabilities P_{ij} and the predissociating decay rate k_r were kept the same as for model I, and the values of $P_{i-1, i+1}$ were assumed to be quite large, of the order of $0.3P_{i-1, i}$, $P_{i, i+1}$, in order to exaggerate any effect of non-nearest-neighbour transitions. The behaviour of the eigenvalues did not change, of course, from the pattern shown in Fig. 2, except in minor numerical detail. Referring only to λ_N , the high-pressure limiting value remained the same as before *i.e.* eq. 2, as de-

manded by the physics of the situation, and the low-pressure limiting value was about 1% higher than before, eq. 3; the latter is reasonable since the predissociating level is now fed by two pathways from below, from levels $i = r - 1$ and $i = r - 2$, and the net result is a slightly higher non-equilibrium population n_r . However, in the intermediate-pressure region, the values of $-\lambda_N$ again conformed, to better than 1 part in 10^6 , to the Lindemann form, eq. 5. Consequently, we can assume from now on that the step-ladder or nearest-neighbour assumption is only a matter of detail; it will affect the calculated rate somewhat, as it also did in the diatomic dissociation case, but it will not lead to qualitatively invalid results: this of course is a great advantage, because matrix [1] is then only a tridiagonal matrix, and once symmetrised, can be manipulated on a computer with great efficiency. An alternative (but not quite rigorous) rationalisation of this is to say that a full symmetric matrix can always be reduced to tridiagonal form: hence, a nearest-neighbour model, given suitable weighted transition rates, will always mimic the behaviour of the full transition matrix.

A brief digression: it is common in most unimolecular reaction theories to make the assumption, for want of something better, of unit deactivation efficiency, at least for the interesting states. Our model I has deactivation efficiencies which are monotonically increasing as the index i increases, but all of them are close to 10^{-3} per collision in magnitude. The calculation was repeated with all downward transitions $P_{i, i+1}$ having the value of unity, which does no more than shift Fig. 2 to the left by three log units. Alternatively if we make some deactivation probabilities unity, and leave others as 10^{-3} , the thickness of the bundle increases by about three log units, and with it, the width of the (shaded) interaction region. Hence, assumptions about the numerical magnitudes of the transition probabilities themselves also only affect the detail of the calculation, but not the overall picture.

Returning now to the question of more than one decay channel, *i.e.* model III. Calculations were performed using the same standard set of P_{ij} , with k_r having its usual value of 10^{13} s^{-1} , and with various values assigned to $k_{r+1}, k_{r+2}, \dots$. The pattern of eigenvalue behaviour remains very much as is shown in Fig. 2, except that there is now one eigenvalue, horizontal in the top left-hand portion of the diagram, for each non-zero

value of k_i , and correspondingly one less in the diagonal bundle. At very low pressures, $\lambda_0, \lambda_1, \lambda_2, \dots$ are exactly the values assigned to the k_i in descending order of magnitude. If all the k_i are the same, e.g. 10^{13} s^{-1} , then there is a set of degenerate λ_j at this value: this degeneracy begins to split as the interaction region is approached, and each one merges into the corresponding member of the bundle (see inset to Fig. 2). At the high-pressure limit, the obvious result was obtained for the rate constant, viz.

$$[6] \quad -\lambda_N = k_{\text{uni}} = \sum_{i \geq r} \tilde{n}_i k_i$$

At the low-pressure limit, a rather startling, but in hindsight very obvious result was obtained, namely that k_{uni} was *exactly* the same as if there was only one decay channel, k_r ! The physical explanation is very simple, and in fact is an implicit assumption of earlier master-equation theories (2, 15): level $i = r$ decays to products just as quickly as molecules can be supplied to it from level $i = r - 1$; consequently, in the limit, no molecules can ever find their way into levels $i = r + 1, r + 2, \dots$, and these decay channels become inoperative at very low pressures.

Moreover, the form of k_{uni} in the intermediate-pressure region can no longer be fitted to the Lindemann form. This is an interesting asymmetry: models I and II have a common high-pressure limit, but different low-pressure limits (arising from additional off-diagonal elements in the matrix); models I and III have a common low-pressure limit, but different high-pressure limits (arising from additional diagonal elements in the matrix); models I and II are Lindemann, model III is not.

Another digression: in order to confirm the interpretation that at the low-pressure limit the steady-state population distribution contains no molecules in states $i > r$, it is necessary to generate the population distribution for a time t in the process for which $|\lambda_j^{-1}| < t < k_{\text{uni}}^{-1}$. This is most easily done using model IV, Fig. 1d, which conserves particles. Model IV is identical with model I except for the addition of one more equation representing the concentration of the (unstructured) product molecules B; the equations now conserve particles, and the generation of the population distribution, in principle at least, follows simply in the normal way (20). The equilibrium concentration of B was chosen to be large (as it is in many real isomerisation experi-

ments e.g. methyl isocyanide, cyclopropane): this makes $\bar{k}_r$ very small, yielding the result that apart from the introduction of the conservative zero eigenvalue, the remaining N eigenvalues λ_j do not change. It turns out that the calculation of the steady-state population distribution is relatively straightforward in the intermediate pressure range, but is rather ill-conditioned at the pressure extremities, especially at low pressure. At relatively high pressures, where λ_{N-1} is within 0.1% the high-pressure limit, we recover the expected result that all levels below $i = r$ are in thermal equilibrium with each other, all levels above $i = r$ are also in equilibrium with each other, but there is a disparity of the order of 0.1% between the two sets of populations. At relatively low pressures, where $[M]^{-1}\lambda_{N-1}$ is (say) about 1% above the limiting value (it is difficult to get much nearer to the low-pressure limit because of severe cancellation problems) the predicted result is also recovered: levels below $i = r$ are in equilibrium, more or less, with each other, but with the degree of equilibration falling successively as i increases towards r , very much like plots given by Tardy and Rabinovitch some years ago (16); n_r is of course about 1% greater at this pressure than the value given by eq. 4, and all the levels above $i = r$ are very closely in equilibrium with each other, having populations of the order of 0.1% of their normal equilibrium values. Thus, the reason for the common low-pressure limit for models I and III is confirmed.

Now, if the behaviour of k_{uni} for model III does not conform to a Lindemann plot, does it have some other recognisable form? It was found that if two non-zero values of k_i were used viz. $k_r = k_{r+1} = 10^{13} \text{ s}^{-1}$, the fall-off behaviour of k_{uni} could be represented very well by a Kassel curve. The conformity to a Kassel curve was not exact, but nowhere was the deviation more than 0.1%, and the best value of s for this case was $s = 1.14$, using, of course, the high-pressure Arrhenius parameters appropriate to this model, i.e. eq. 6. The most interesting feature of this observation, however, is that at the temperature in question, the rate constant given by eq. 6 was just 1.14 times the value of $k_r \tilde{n}_r$, the single-channel value. Let us therefore introduce the conjecture that

$$[7] \quad s \cong \sigma = \sum_{i \geq r} \tilde{n}_i k_i / \tilde{n}_r k_r ?$$

Two model calculations were then constructed

having the value of $\sigma = 2$. This can be done for example by keeping both k_r and k_{r+1} fixed at 10^{13} s^{-1} and increasing the degeneracy g_{r+1} to give the desired result, or by keeping the degeneracy fixed and increasing k_{r+1} correspondingly; either way, the best fit to a Kassel curve was obtained when s was within the range 1.98 to 2.02. The coincidence between s and σ is remarkable and suggestive: however, an attempt to extend eq. 7 to higher values of σ does not lead to a one-to-one correspondence between s and σ ; for example, two calculations with $\sigma = 3$ (one with equal contributions to the rate from $i = r, r + 1$, and $r + 2$, and the other using only two channels $i = r$ and $r + 1$) both gave good fits to Kassel curves having $s \approx 2.65$. Considering the density of states in the critical region and the relatively narrow width of the bundle of internal eigenvalues, the calculations for $\sigma = 3$ really are very strained and probably unrealistic, and it would seem therefore not unreasonable at the present juncture to leave conjecture [7] as it stands, to be verified for larger σ at a later stage using much larger matrices.

Relationship to Existing Theories

Rather than compare the present approach specifically and in turn with each of Hinshelwood-Lindemann theory (2), Slater theory (2, 21), and RRK or RRKM theory (2, 3), it seems more economical at this primitive state of development to make associations with the general body of ideas used in these theories, and highlight the shortcomings where they appear. We have constructed an incredibly primitive master-equation formulation of a unimolecular reaction. The fact that this is worth attempting stems from our experience with the master-equation treatment of diatomic dissociation, (a) that the step-ladder activation assumption will give results only quantitatively but not qualitatively different from a full network of activation steps, and (b) that once the network becomes sufficiently large, the reaction rate is rather insensitive to the internal transition rates. The molecule is represented only as a series of energy levels: they are not subdivided into classes or subgroups which could be regarded as oscillators in the normal sense, and yet fall-off behaviour is found which is often treated using the idea of a number s of coupled oscillators. In a sense, the present model corresponds to complete randomisation of energy and could easily be extended, in principle at least, to

accommodate separate interleaving manifolds of energy levels with high transition rates within each manifold and relatively low transition rates between members of different manifolds to examine effects caused by non-randomisation of energy. The concept of oscillators in other theories is associated with both the *shape* of the fall-off curve and the *decline* of the Arrhenius temperature coefficient with pressure; in practice also, there is a correspondence between the complexity of the molecule and the *position* of the fall-off curve on the pressure scale. We have shown that the varying shape of the fall-off curve may have an alternative interpretation: if we use the Kassel s to describe approximately the shape of the fall-off curve, then s appears via the conjecture [7] to be the ratio at infinite pressure of the total rate to the contribution of that particular decay channel which persists at the low-pressure limit.

On the other hand, there is no direct correspondence between the value of σ (or s) and the position of the fall-off. As we showed, the position of the fall-off is adjustable trivially by changing the mean value of the deactivation probabilities in the system. The major determinant of the position of the fall-off curve on the $\log [M]$ axis is simply that the top left-hand portion of the interaction region (*cf.* Fig. 2) occurs when

$$[8] \quad k_r = -\lambda_0' \sim [M]ZP_{ij}(\text{max})$$

where λ_0' is the eigenvalue of largest numerical magnitude for the unperturbed matrix (*i.e.* all $k_i = 0$), and $P_{ij}(\text{max})$ is the largest collisional deactivation probability in the system. Thereafter, each successive member of the bundle of eigenvalues exhibits its displacement at a successively higher pressure, and therefore k_{uni} bends over and becomes horizontal at a pressure value higher than that given in [8] by an amount determined in some topological fashion by the *thickness* of the bundle; thus, eq. 8 is the major determinant of the position of the fall-off, with the thickness of the bundle being a secondary factor. Clearly there is some scope for establishing a tenuous connection between σ and the thickness of the bundle of eigenvalues corresponding to the internal processes: this aspect has not been examined in detail, but it seems obvious that the main factor will be the dynamic range of deactivation probabilities for the dominant steps in the network, and this range will probably be greater

for more complex molecules than for simple ones. The necessary relationship which we cannot establish is the one between σ and the k_i required to define the position of the interaction region via eq. 8; this of course is where oscillator-based theories have some strength since, for a fixed deactivation probability, the predicted fall-off pressure decreases with increasing complexity of the molecule.

The present model yields distinctly different Arrhenius temperature coefficients at the high- and low-pressure limits. At the low-pressure limit, when the reaction flux is through only the $i = r$ channel (in the single-step excitation model) or only a few channels just above $i = r$ (in the general multi-step excitation model), the behaviour of the rate constant, to a first approximation, is $\sim \exp(-\epsilon_r/kT)$. Another factor will be the way in which the depopulation of the predissociating levels and of those below them change with temperature, which will depend to a large extent on the temperature coefficients of the internal relaxation eigenvalues. Making the plausible assumption that any non-equilibrium effects become more severe at higher temperatures, the Arrhenius temperature coefficient would then be just slightly less than ϵ_r in numerical magnitude (4). At the high-pressure limit, where the rate constant behaves as $\sigma g_r \exp(-\epsilon_r/kT)$ (from eq. 7), the Arrhenius temperature coefficient will depend on ϵ_r again, and on the way σ varies with temperature. If σ is small, there will be relatively little scope for deviation from ϵ_r , if σ is large, there may be considerable scope, so in a sense we recover a relationship between the change in Arrhenius temperature coefficient with pressure and σ (crudely the number of effective oscillators), but not a quantitative one as in other theories. We may distinguish two types of behaviour, one which we may regard as 'normal' and one 'abnormal'. (a) If reaction takes place from states immediately above the state $i = r$, and the products $k_i g_i$ for the dominant reactive channels are not very different from the product $k_r g_r$ for the critical level, then the 'activation energy' will be only a little in excess of ϵ_r , and the frequency factor A_∞ will be of the order $\bar{k}_i g_i$. (b) If, because of a peculiar distribution of predissociating rotational states of high J somewhat above the level $i = r$, with $\bar{k}_i g_i \gg k_r g_r$, one can have an Arrhenius temperature coefficient considerably greater than ϵ_r , and a 'high' frequency factor A_∞ .

The behaviour of these two cases as the pressure decreases will be different: case (a) will exhibit a small decline in 'activation energy' with a relatively constant frequency factor, whereas case (b) will exhibit a large decline in 'activation energy' and also a relatively severe change in frequency factor. Obviously σ plays some role in the change in both cases, but not a clearly defined one.

In discussing the results of a similar investigation about 10 years ago, Valance and Schlag (15) concluded that there were three necessary and sufficient conditions for RRK behaviour of the master-equation model: (i) the steady-state assumption; (ii) the strong-collision assumption; (iii) all levels below the critical energy are required to be in equilibrium. We have not established in a detailed sense that the present model gives exactly RRK behaviour, only that it is Kassel-like or Slater-like, but without *any* of these artificial restrictions, which it would seem arise from the partitioning procedure used by Valance and Schlag rather than the physics of the situation: essentially, this partitioning reduced the problem to a two- or three-state problem (admittedly with internal structure within the two or three states concerned) and the risks attendant upon making such drastic assumptions are by now reasonably well recognised (4).

Finally we note that, of course, one of the attractions of the master-equation approach is that it can encompass naturally different efficiencies of colliding molecules in restoring the fallen-off rate, which is not always easy in other approaches. It does not seem entirely unreasonable to envisage at this stage a hybrid theory in which the details of the internal relaxations are treated in the present fashion but the calculation of the lifetimes of the interesting states rests heavily on RRKM-like considerations.

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